NexTimber® by Timberlink

IT'S WHAT BETTER TOMORROWS ARE BUILT ON

DfMA Guide

Version 1.0
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Contents

Introduction

Purpose of this Document

This Design for Manufacture and Assembly (DfMA) Guide:

- is for Designers working with NeXTimber® Cross-Laminated Timber (CLT) and Glue-Laminated Timber (GLT) products
- provides guidance on the manufacturing process using CNC machinery for NeXTimber® CLT and GLT products, and how manufacturing or overall project time and cost may be impacted by the design choices of the Designers in the Design Documentation
- is intended to assist Designers in optimising the Design Documentation by taking into account the factors and examples set out in this guide

By adopting a holistic approach that integrates this DfMA Guide's principles throughout stages of the project lifecycle, Designers can maximise efficiency, minimise waste, and unlock the full potential of timber as a versatile and sustainable building material.

However, it's important to note that while there are certain aspects within NeXTimber's control during the manufacturing phase, decisions made by Designers during the design phase have a significant influence on both the manufacturing and the overall project. Design choices related to component sizing, connection details, and assembly sequences significantly impact the manufacturability and constructability of the final structure.

Every construction project is unique, with its own set of challenges, constraints, and opportunities. Therefore, while this guide offers valuable insight into the manufacturing process, Designers need to exercise judgment and professional expertise to adapt that insight to suit the specific requirements of their projects.

Any input by NeXTimber® during any collaboration or consultation with Designers is subject to the NeXTimber® Terms and Conditions of Supply.

Unless otherwise defined in this guide, defined terms have the same meaning as in the NeXTimber® Terms and Conditions of Supply.

Glossary of Terms

Billet	Large format, unprocessed, CLT element from which individual panels are machined.
CLT	Cross-Laminated Timber (panels)
CNC	Computer Numerical Control (timber processing machine)
Designers and/or Design Team	The design team contracted by the customer and typically comprises architects, engineers, building designers, mechanical engineers and other design professionals contributing to the specification of NeXTimber® products.
Design Documentation	Documents prepared by the engineer, architect or other consultants necessary to complete NeXTimber® deliverables
DfMA	Design for Manufacturing and Assembly
GLT	Glue-Laminated Timber (beams and columns)
Manufacturing Documentation	Digital documentation produced by NeXTimber®, required to complete the manufacturing of the project i.e. the 3D digital model, CNC files, QA documentation
NeXTimber® Terms and Conditions of Supply	The terms and conditions of the contract on which NeXTimber® will supply the NeXTimber® CLT and GLT products.
Panel	Components that are cut or machined from a Billet
QA	Quality Assurance

What is DfMA?

Design for Manufacture and Assembly (DfMA) is a philosophy that emphasises optimising the design for ease of manufacture and assembly. In the context of mass timber construction, DfMA involves designing and detailing timber components fabricated off-site in a manner that rationalises component count, minimises waste and streamlines assembly processes to reduce onsite assembly program with the intent of reducing overall project costs.

By integrating DfMA principles, mass timber construction can capitalise on the benefits of off-site fabrication, such as improved precision and accuracy, reduced site labour and disruption and enhanced sustainability.

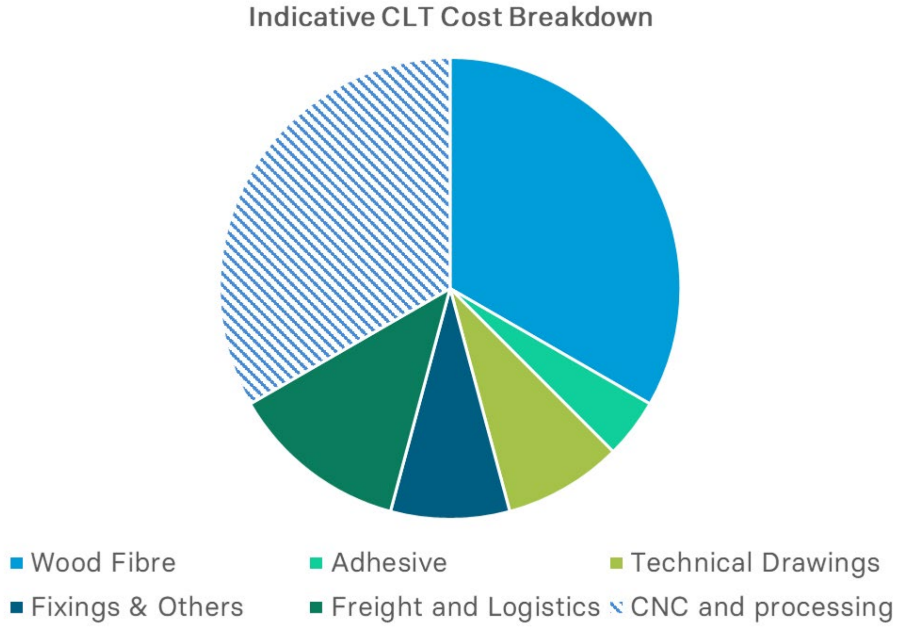
Designers taking the time to understand the constraints of the NeXTimber® manufacturing process and incorporate them into their Design Documentation can offer cost benefits to the project with regard to CLT production and delivery.

Competing Interests

There is not a one-size-fits-all approach to DfMA with timber buildings. There are often competing interests between manufacturing and installation efficiencies which should be assessed and resolved as early as possible.

Timber manufacturers generally will aim to prioritise what is cost-effective for them to manufacture, and conversely, contractors will aim to prioritise what is easy for them to erect and install. Taking these interests into account, a holistic approach considering the best outcome for the project generally yields favourable outcomes. During the design stage, consider what design approach and details are best for the project, taking into account that different requirements may be prioritised depending on the project's needs, timelines and budget and allow Designers to prioritise their focus.

To be best equipped to make some of these decisions, it is advantageous to get an understanding of the manufacturing process, what information is required by NeXTimber®, what its manufacturing capabilities and constraints are, and appreciate that minimising material use may not be the most cost-effective design solution.



Fundamentally, the cost of manufacturing CLT panels can vary significantly depending on the design complexity and additional labour costs. The single biggest unknown with an underdeveloped design is the time and costs associated with the processing. These unknowns can lead to variances and uncertainty in the costing, time to produce as well as the overall delivery schedule.

The other factors such as fibre, adhesive, freight and to a lesser degree, fixings, tend to be far more quantifiable at an early design stage and tend to have the least level of variability.

Digital Technology and Documentation

Clear and detailed documentation plays an important role in the success of DfMA for several reasons:

Clarity in Communication: Detailed documentation ensures that parties involved in the project, from Designers to manufacturers to assemblers, understand the requirements and specifications clearly. This minimises the risk of misinterpretation or errors during the manufacturing and assembly stages, particularly with regards to screw and connection placement and should clearly be shown on the engineering documentation.

Efficient Manufacturing: Clear documentation enables manufacturers to understand the design intent accurately, leading to more efficient production processes. This reduces the likelihood of rework, delays, or costly mistakes during manufacturing.

Streamlined Assembly: Detailed documentation provides assembly teams with clear instructions on how components fit together, reducing the time and effort required for assembly. This can result in faster assembly times, lower labour costs, and improved overall productivity.

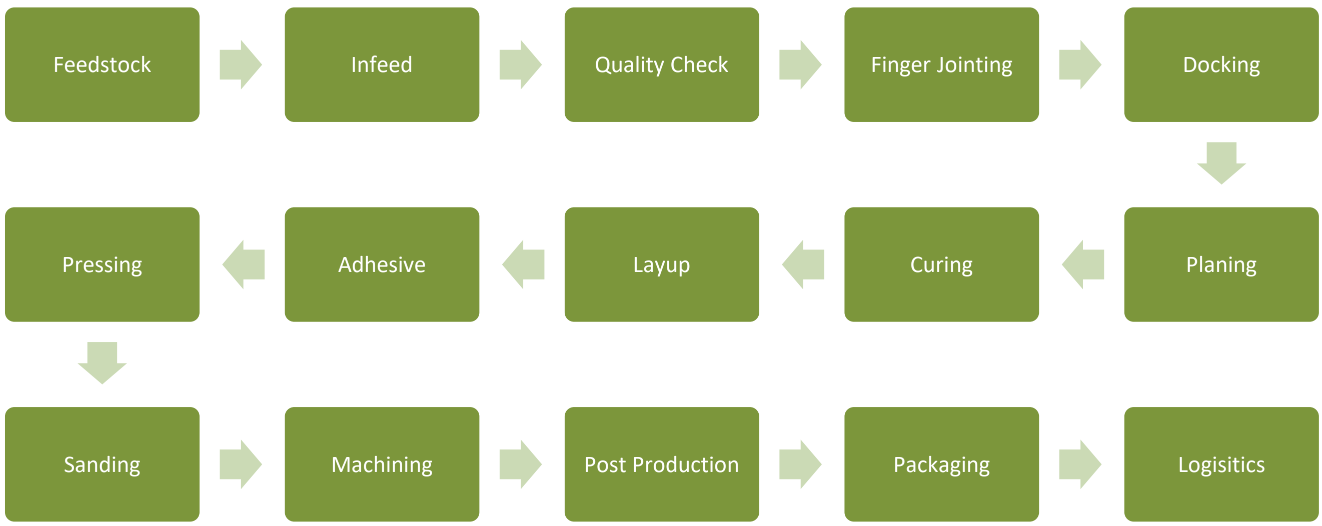
Responsibility of Scope

Clarify the scope of each party within the Design Team (note this may vary depending on the procurement model, size of project and scope of works agreed with NeXTimber®), and clearly determine which party is responsible for providing the documentation required to complete the site assembly – namely:

- panel layout and assembly plans
- fixings take-off and procurement
- site connection set out plans
- panel sequencing
- loading and despatch.

Typical NeXTimber® CLT Manufacturing Process

The CLT and GLT manufacturing process is linear and follows the below process. There is little to no manual intervention in the production process.



Billets and Panels

A "Billet" refers to the stacked and glued layers of structurally graded radiata pine timber, alternating layers laminated at right angles to form a solid homogenous timber element. The Billets typically comprise three, five, seven or nine layers.

Panels are then "nested" into billets - meaning that they are digitally placed for cutting, trimming and finishing. Individual panels are cut from a Billet, and depending on the size of the components, a Billet may comprise one or several panels.

Essentially, a CLT "panel" is the finished product. A pressed blank is a "Billet".

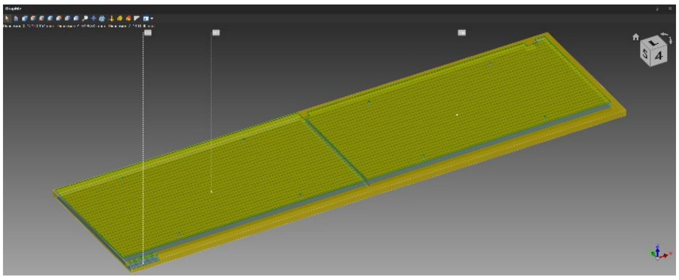


Image: Nested panels inside a billet

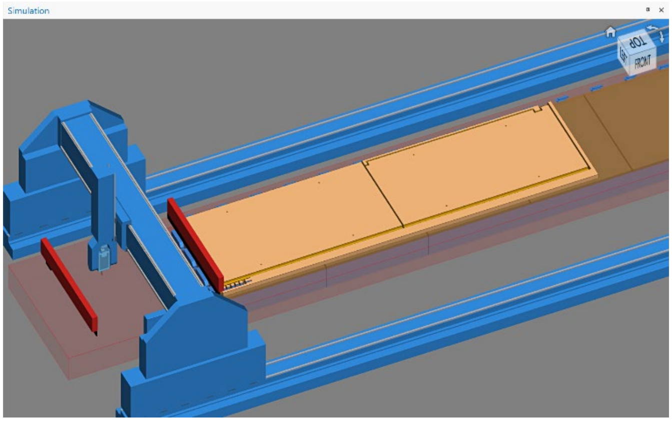


Image: Nested panels inside a billet for CNC processing simulation.

NeXTimber® Billet Manufacturing Capabilities

Minimum – maximum length	6m to 16m
Minimum – maximum width	2.4m to 3.5m
Minimum - maximum thickness	90mm to 360mm

Bear in mind that there are certain 'dead spots' within the above ranges that can produce a significant amount of offcuts when nesting panels within the Billets. For example, given the geometric constraints for Billet production, panels less than 2.4m wide, and also for multiples of panels between 1.7m and 2.4m in width, can fall into an awkward dimensional range that generates inefficient nesting patterns. Subsequently, a reasonable amount of offcut is produced – increasing the quantity of fibre and overall cost. Similar principles apply for length.

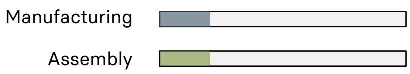
Given that NeXTimber® aims to manufacture on a project-by-project basis, we tend not to keep the offcuts and the project is charged according to how many Billets are needed to be manufactured to produce the project, so overlooking this may contribute to unexpected material loss and costs.

Factory Handling

Handling is done within the factory using an overhead vacuum crane fabric slings and/or forklifts.

Indicative Complexity Legend

Throughout this guide, the following legends are (either both or single bars) provided to help illustrate the estimated indicative level of complexity from both a manufacturing and assembly perspective. Note that they do not take into account unforeseen challenges that may arise as a result of component size, site access and nesting complexities as encountered on a project-by-project basis.



Design for Manufacturing

Fabrication and Machining

When employing a CNC (Computer Numerical Control) machine to process mass timber elements with a DfMA mindset, accuracy and machining efficiency are key considerations. NeXTimber® generates detailed digital models of the elements including the connections and joints, translating the Design Documentation into a digital model, incorporating the precise dimensions and geometries required for each component. These digital models are then translated into data for the production (Project Data) of the Billets as well as the files for CNC machining.

The use of CNC machining enables repeatability and consistency across multiple components, contributing to the overall quality and uniformity of the construction process. Although CNC technology can cut intricate and highly accurate details, it doesn't necessarily mean that this is best practice given the impact that it can have on factory throughput should processing time slow down production.

However, by optimising the design for CNC fabrication, Designers can leverage the benefits of DfMA, resulting in faster production and construction timelines, reduced manufacturing waste, and improved overall project efficiency.

Component Size

In many cases, having bigger and fewer timber components can be advantageous to DfMA principles in mass timber construction. Namely through:

Simplified Manufacturing: Larger components require fewer individual pieces to be manufactured, reducing the number of manufacturing steps and potential points of failure. This simplifies the manufacturing process, streamlining production and improving efficiency.

Reduced Assembly Complexity: With fewer components, assembly becomes less complex and time-consuming. Larger components often require fewer connections and joints, resulting in faster assembly times and reduced labour costs.

Enhanced Structural Integrity: Larger timber components can provide greater structural stability and load-bearing capacity compared to smaller components. This can result in a more robust and resilient building structure overall.

Ease of Transportation and Handling: Transporting and handling fewer, larger components is generally more straightforward and cost-effective than dealing with numerous smaller pieces. It reduces transportation logistics and on-site storage requirements, minimising potential delays and complications.

Quality Control: Managing quality control is generally easier with larger components, as there are fewer pieces to inspect and monitor during the manufacturing process. This can lead to improved consistency and reliability in the final product.

However, there are also considerations where smaller components may be preferred, such as:

Site Constraints: In cases where access to the construction site is limited or where there are spatial constraints, smaller components may be more practical as they can be easier to manoeuvre and install in confined spaces.

Design Flexibility: Smaller components offer greater flexibility in design, allowing for more intricate and customisable structures. This can be advantageous in projects where architectural complexity or uniqueness is a priority.

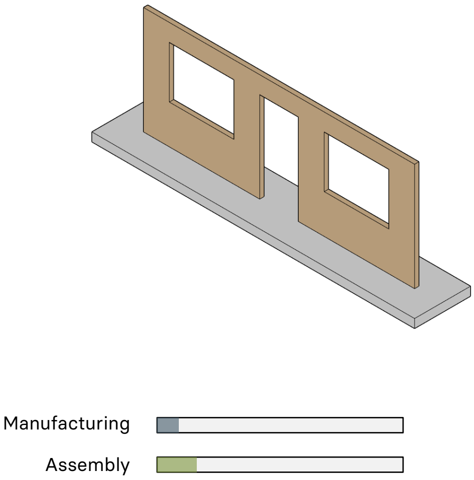
Material Optimisation: Smaller components can help optimise material usage by minimising waste and allowing for more efficient utilisation of timber resources.

Ultimately, the decision around component size depends on various factors, including project requirements, site conditions, budget constraints, and general design considerations. Striking the right balance between these factors is essential to achieving the most efficient and effective construction outcome while adhering to DfMA principles.

Small Panels Versus Large Format Panels

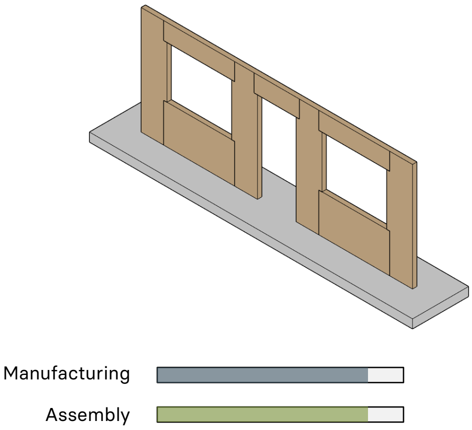
The design of a CLT wall, as shown below, illustrates some of the fundamental considerations when panelising or breaking up a wall elevation. There may be instances when a wall needs to be broken into a number of smaller elements (ie. structural/articulation requirements, size and weight of panel limitations), but generally speaking the larger format panels are typically more efficient in a holistic sense.

Single CLT panel with CNC Hole Cut-Outs



- Minimal CNC machining time
- Cut-outs create material waste
- More difficult to manoeuvre within factory
- Minimal number of site lifts / crane hook time
- Minimal number of on-site connections (less labour)

Multiple CLT panels - Site fixed



- Adds to Billet complexity and requires a higher level of CNC machining time. See following.
- Reduced waste / more efficient nesting (no cut-outs)
- Greater co-ordination and QA process required
- Greater level of structural articulation
- Higher number of site lifts / crane hook time
- High number of on-site connections (more labour)
- Greater amount of fixings (material cost)
- Greater amount of temporary propping and bracing
- Smaller and lighter components may be better suited to site constraints

Note that the final design solution may be a hybrid of the above two scenarios.

The reason that this matters to NeXTimber® is due to factory throughput. As the production line is continuous, the production line may need to slow down to accommodate the billets that are slow in the CNC and/or panels in the post-production process – ultimately reducing the overall production capacity of the factory and attracting additional labour resources (handling, packaging, QA) and potentially adding to lost opportunity.

Standardised Panel Mix

To increase factory productivity, it is preferable to standardise the panel mix across the project. For example, avoid creating a patchwork of different Billets, as there is a greater opportunity to maximise recovery from the Billet once the panels are nested for processing.

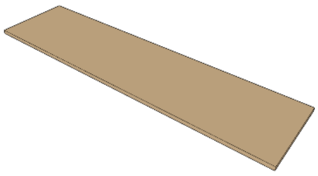
Rationalising the types of panels allows NeXTimber® to optimise productivity by:

- Optimising panel nesting within panels and reducing unnecessary residues
- Reducing the number of overall Billets
- Optimising feedstock management through the production line
- Optimises QA sampling rate and testing

Panel Nesting and Machining Complexity

Very Low

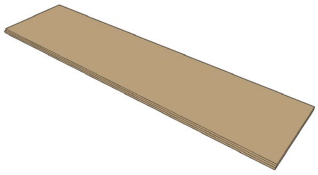
Square cut to perimeter



Manufacturing

Low

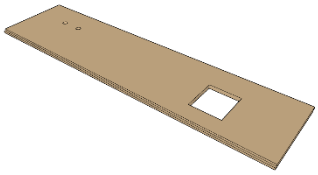
Square cuts, rebates to long edges



Manufacturing

Moderate

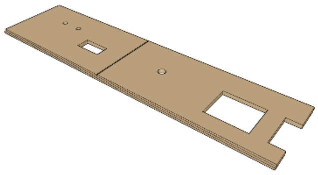
Additional openings and penetrations



Manufacturing

High

Additional chases, rebates, openings and penetrations

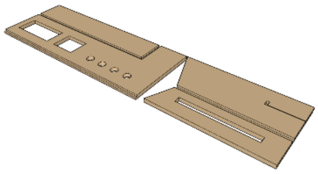


Manufacturing

Very High

Small panels with rebates, chases, openings and drillings, intricate machine work, squared corners and QC checks.

Added logistics complexity.



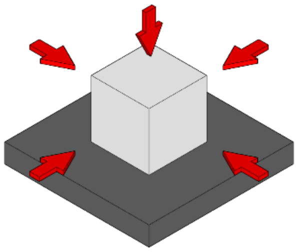
Manufacturing

Post-Production Manufacturing

Machining Capabilities

NeXTimber® utilise a 5 Axis Hundegger PBA Industry, a German-manufactured gantry style CNC.

5 axis means that the panel is accessible from the top, left, right, front and back, but not from the underside. When processing is required on the underside of the panel or if there is intricate processing on the edge of the panel, then the panel may need to be flipped on the CNC to finish the processing.



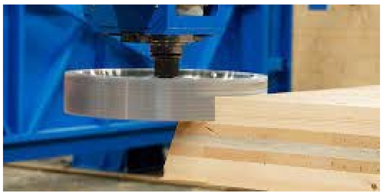
Tooling

1100mm (8.5mm kerf) 5 axis circular saw

5 axis chainsaw (10mm chain width)

20, 40 and 80mm diameter finger mills / routers

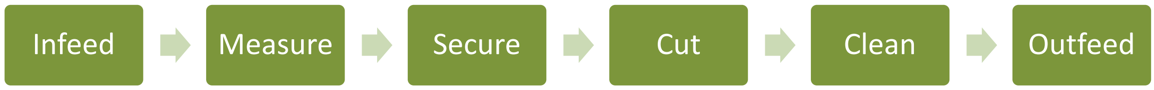
40mm deep and 100mm deep disc mills



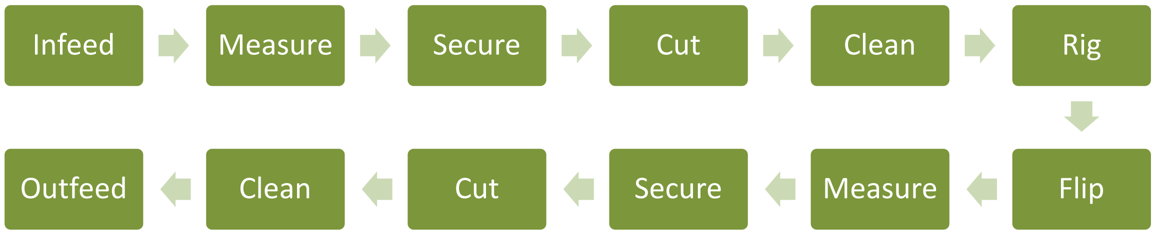
Cornering tool and various drill sizes

Although various tools are available to perform processes, some tools are more efficient than others. Minimising tool changes to process the panel can increase the panel throughput and productivity and hence speed of manufacture.

Standard Cutting Process



Standard Cutting Process for Panel Flipping



The reason that this is important to NeXTimber® is that the process of flipping panels requires labour resources, increases processing time significantly and slows factory throughput.

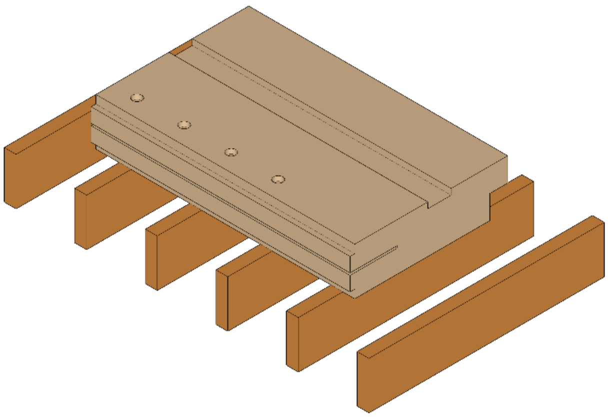
When a panel needs to be flipped to be machined, it effectively needs to be processed twice – plus some, and re-orientating a panel takes anywhere between 20mins and 30mins which reduces productivity and consumes manufacturing opportunity and increases project costs.

Single Side Processing

To maintain manufacturing efficiencies, it is advisable to consider detailing that requires a single side operation and reduced handling complexity. Below are some examples for illustrative purposes.

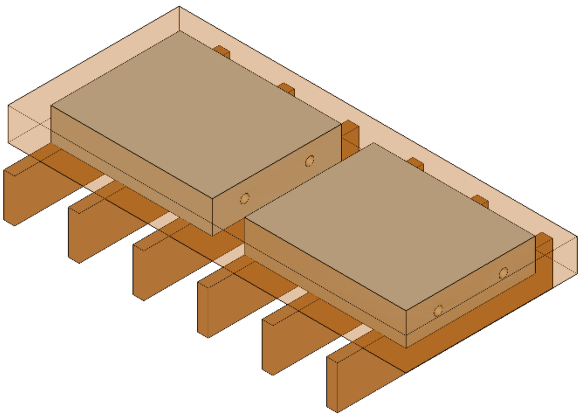
Machining finished surface side, or external face upwards, is also desirable as to streamline the stacking and loading. This also improves productivity if temporary membranes are to be factory applied.

Although processing upside down may simplify the machine operations, the panel may still require a panel flip to re-orientate prior to despatch.

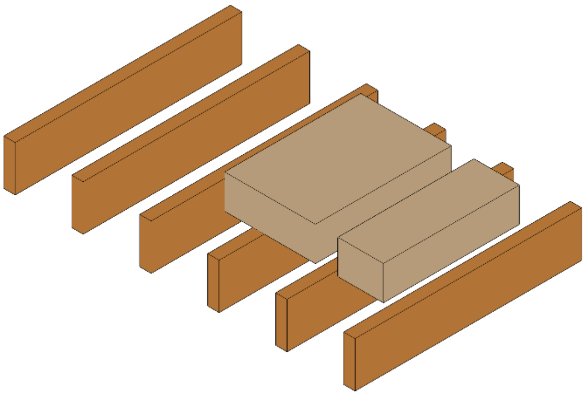


In most instances, deep laps (greater than 40mm in depth) can be processed on the underside of the panel.

- Drillings, trenches and counterbores need to be performed from top surfaces.
- Intricate and shallow details need to be cut from top surface due to the size and diameters of tools. Some larger rebates can be undercut.
- This panel would require processing from both sides due to the narrow rebate on the front edge.



Where panels need to be nested within a Billet and processing on ends, it may be necessary to remove a panel to complete the tooling operations, requiring the CNC to stop and start while waste and material is removed.



Panels are supported on timber bearers for CNC processing. Small or narrow elements can become unstable during the processing and fall between the bearers. Where possible, these should be nested so that they can be processed over at least three bearers, which may increase material. Ideally, small or narrow parts are avoided.

Machining Inclusions

NeXTimber® CLT and GLT products are manufactured, combining the efficiencies of digital technology and the accuracies of CNC capabilities. NeXTimber® products come pre-cut to finished dimensions including all windows and door openings, rebates, joints and lifting points.

Standard Machining Examples

- Panel-to-panel joints and rebates
- Door and window openings
- Square rebates for beams to wall connections

Custom / Special Machining Examples

- Rebates and channels for custom steel bracketry and components
- Special angled and mitre cuts
- Intricate machining for embedded steelwork
- Light socket and power point rebates and channels
- Rebates and trenches for building services

Included in our supply offer is a standard level of CNC machining and more extensive, custom machining can be provided at an additional cost and as required on a project-specific basis.

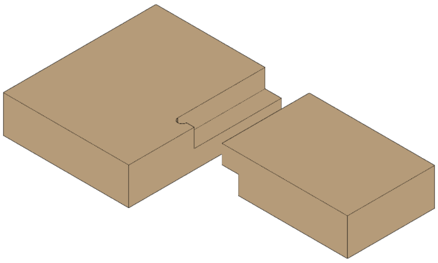
Rounded Corners

It is common practice to use routers to cut corners for openings (windows, doors, service penetrations) as they cut vertically and access full depth of cut unlike a circular saw's limitations due to the large radius of the blade.

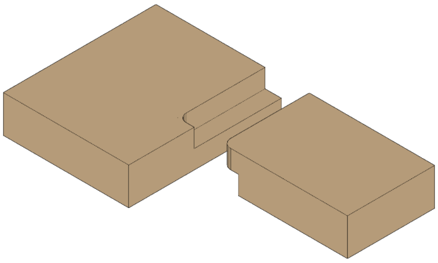
- Where necessary, corners can be corrected either:
    - manually cut by hand in post-processing; or
    - using a specifically designed corner cutter tool.

Both of the above are separate tool operations and require a tool change and separate process. Where possible, consider using rounded corners, or overcuts where the corner will not be visible or aesthetically and structurally acceptable.

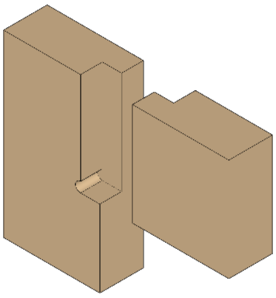
Overcut lap joint



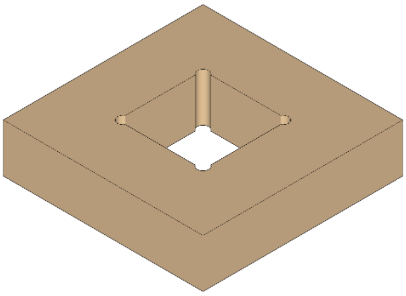
Rounded lap joint



Overcut lap joint (lintel)



Overcut penetration



- Ideally, where square corners are absolutely necessary, these should be nominated on the Design Documentation. By default, it will be assumed by NeXTimber® that overcutting and rounded corners are permissible. Squared-off corners will attract an additional charge.
- Structural detailing should show the intent of the likely process that will be required to process the detail. The final operation will be determined during the development of the manufacturing model.

Design for Assembly

In mass timber construction, site productivity can be increased by utilising prefabricated mass timber components by detailing and fabricating timber components (while aligning with DfMA principles) with the view of:

Site Safety: Opportunity to pre-fit handrails, and fall arrest systems to reduce live edges, as well as retaining parts of panels around voids such as lifts and facades to act as fall protection.

Manual Handling: Minimising double handling and disruptions to supply of materials by optimising transport, delivery, and storage strategy to suit any site-specific constraints.

Cranage: Detailing connections to reduce crane hook time and lease costs through reduced number of cranage lifts and/or cranage size, whilst maximising utilisation and sequencing. Use of bearing connections where possible, which can be fitted off quickly and accurately, in-lieu of requiring bolts and screws to be fitted off whilst the element is still on the hook.

Site Labor: Reduce site labour costs and install time by minimising the amount of manual fixings to be installed between components. Through precision, prefabricated components machined with CNC-cut connections can be quickly and accurately assembled using minimal tools and labour. This reduces construction time and costs while maintaining high levels of precision.

Off-site Fabrication: Timber components can be precisely manufactured off-site according to the digital models. This allows for standardised production and quality control in a controlled environment, reducing errors and waste. Where possible and practical, pre-fit connections to minimise double handling and the need to manually handle and manage fixings on-site or at height.

Overall, integrating prefabricated connections and joints into the construction and assembly process of mass timber projects enhances efficiency, accuracy, and structural integrity, while aligning with the goals of DfMA to streamline production and improve project outcomes.

Fixings and Connection Selection

Assembly of mass timber buildings will generally comprise a combination of brackets and a variety of screws in different lengths, sizes and thread types. Having an extensive variety of differing screw types on a project can lead to confusion and errors as well as challenges with inventory control. To minimise errors, it is generally good practice to consider:

- rationalising the fixing specification (i.e. larger screws (within reason) and greater spacings);
- avoid multiple screw lengths with small length variations; and
- fully threaded and partially threaded screws of differing lengths and diameter to avoid confusion,

which in turn:

- reduces the likelihood of using the wrong screw in the wrong application;
- minimises stock handling and management and simplifies the procurement process; and
- when conducting multi-level construction, supplies can easily split between floors.

Utilising standardised fixings helps streamline production, reduce errors, and expedite assembly, ultimately enhancing the overall efficiency and quality of the construction process. Additionally, incorporating fixings that allow for disassembly and reassembly can facilitate future modifications or renovations, contributing to the long-term sustainability of the structure.

Connections and Detailing

Tolerances and Details

Methods of connections of steelwork to concrete elements are beyond the scope of this guide. However, the fixing of these elements should consider the different tolerances between the elements. Typically, timber elements are manufactured to far tighter tolerances than other building materials. Having tolerance, or an ability to account for these differences, should be considered as part of the design process. For example:

- It is good practice to oversize shelf angles, taking into account tolerance as well as edge distance to fixings. Make allowance for bolt heads, washers and sufficient thread run out when specifying the horizontal leg of the angle to avoid having to coordinate and notch the CLT panels;
- Consider using cam-style fixing plates for bolted connection to give a greater level of tolerance and accuracy when installing shelf angles to support CLT
- Site-fitted cleats should either allow for horizontally slotted holes, consider long-term whether these need to be locked up, or have the ability to be site surveyed and then site welded for accuracy to avoid having to modify timber components
- Ensure that the proposed fire-rated details can articulate should tolerance and movement be present in the joints and connections

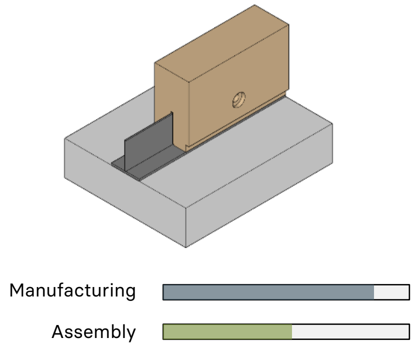
Note: There may be potential additional steel and/or fabrication and detailing costs. However, these costs may often be offset by site labour time savings as well as the cost to coordinate timber, detail and install the timber plate alternatives.

Structural engineering documentation should demonstrate the strategy and approach around tolerances, interfaces, and clearances to other building materials.

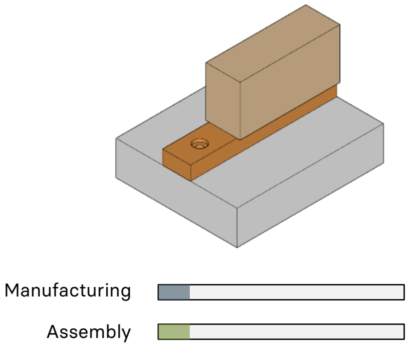
As NeXTimber® manufactures ahead of time (in most instances before the construction of in-situ elements such as concrete and masonry are constructed) allowing for site tolerances within structural details is essential. Alternatively, prescribing stricter tolerances (i.e. verticality and placement of concrete walls) than what is conventionally acceptable may be necessary, provided that they are practically achievable.

Intricate Details - Examples

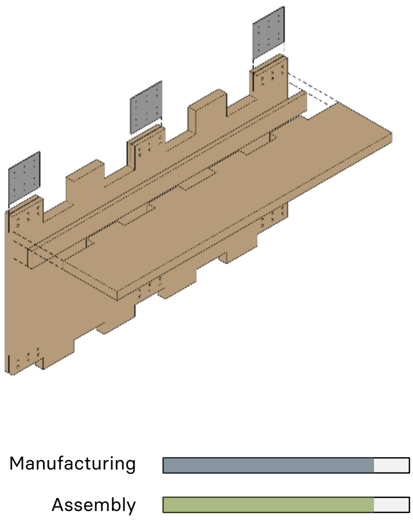
Consider simpler details that may achieve similar architectural outcomes, with less tool and machining operations



- Many tool operations
- 7 separate processes
- Requires processing on both sides of panel

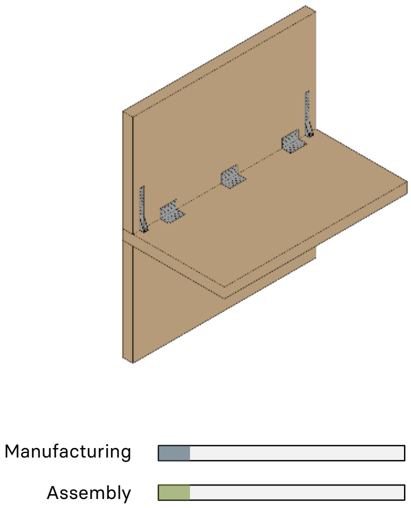


- 1 tool operation
- Processing from single face of panel



Many tool operations, for example:

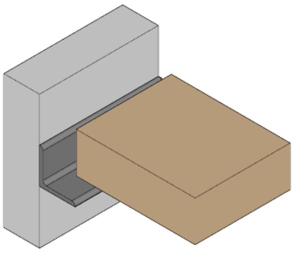
- Cuts, slots and drillings
- Panel flip
- Post processing squaring of corners
- Heavy on documentation and QA



- Simple processes
- Proprietary brackets and fixings
- Floor reinforcement can be added through other means to transmit axial loads

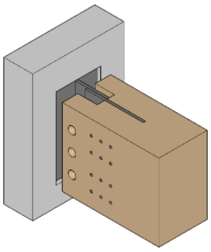
Connections to Concrete - Assembly

Cast in situ concrete elements can vary significantly in terms of flatness and position. When detailing timber connections, it is suggested to consider different material tolerances and known datums and that these are considered in the detailing and documentation so when the components arrive on site they interface with each other and do not delay installation due to elements not fitting together as expected.



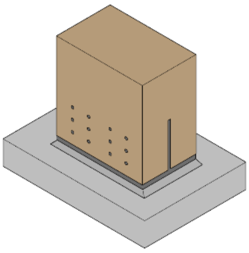
CLT Panel on Steel Shelf Angle

- Build tolerance into the bearing area, ensuring edge distance to fixings can still be achieved
- Make sufficient allowance for the radius of steelwork, bolt heads, washers and thread run out etc.



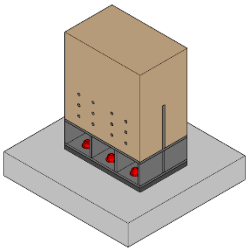
Glulam to extended steel cleat

- Build tolerance into the detail where surfaces can be surveyed and corrections made before the fitment of timber components
- Use steel-to-steel bolted connections where possible to reduce site part count and speed up assembly
- Design so pre-fitment of connections is possible



Glulam column to 'T' base plate

- Build tolerance into the detail where surfaces can be surveyed prior to the fitment of timber components

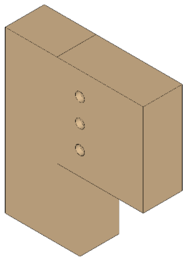


Glulam column to 'T' base plate

- Consider opportunities to pre-fit column baseplates

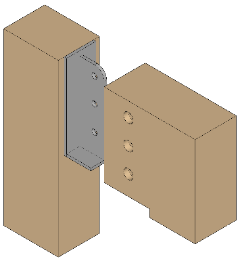
Glulam Connections - Assembly

When detailing glulam connections, it is suggested to consider which parts can be assembled either on the ground (or in a factory) and then lifted into position. By utilising a combination of timber-to-steel, and steel-to-steel connections, part of the assembly can be pre-fitted and quickly bolted together in place – increasing the productivity of the site crane.



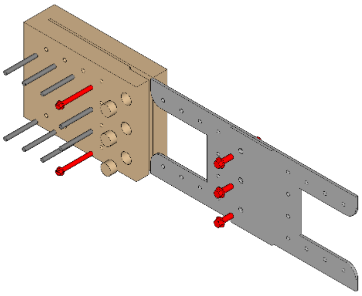
Glulam beam to colmn bearing connection

- Use timber-to-timber bearing connections



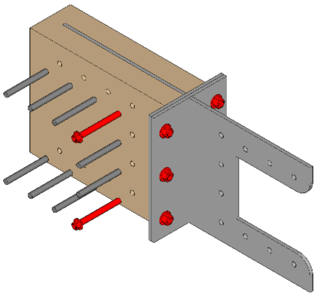
Glulam beam to colmn bearing connection (steel bracket)

- Consider transportation implications if steelwork is to be pre-fitted



Glulam beam splice connection (site bolted)

- Combine timber connections with bolted connections to improve crane utilisation and productivity



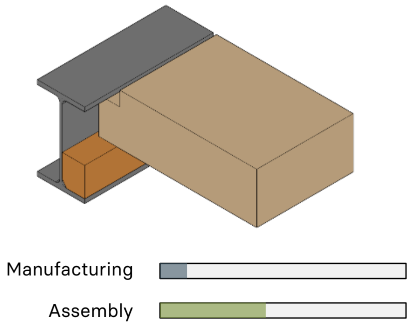
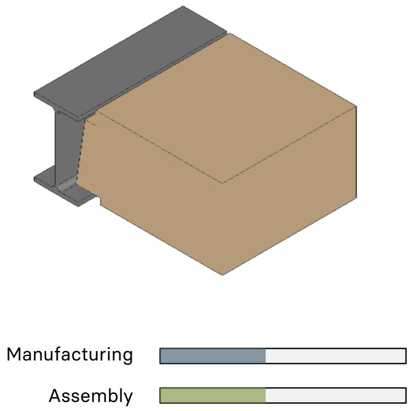
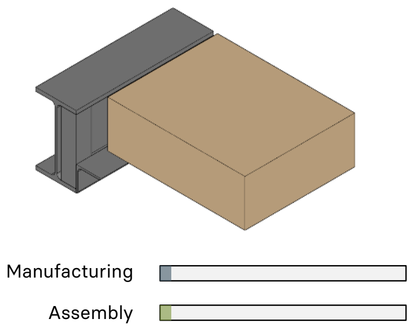
Glulam beam splice connection (site bolted)

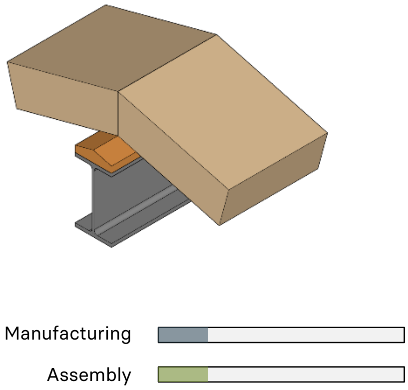
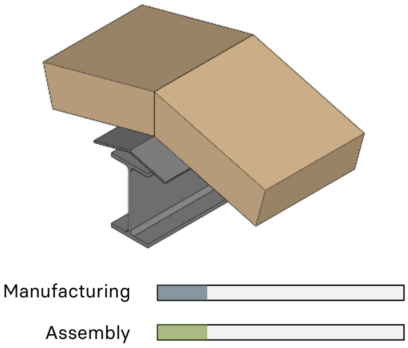
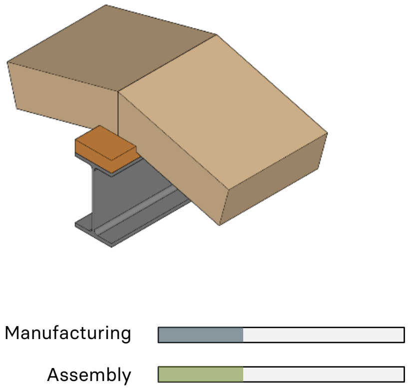
- Combine timber connections with bolted connections to improve crane utilisation and productivity

Interfacing with Steelwork

Connecting steelwork to timber elements and simplifying these connections from a holistic assembly perspective are often overlooked. If timber blocking is necessary, and this needs to be profiled, consider who is procuring it and who is installing it as it may add some unanticipated complexity to the project delivery. Some other things to consider:

- Where steelwork sits within the plane of the CLT, consider the following design options and challenges for the various stakeholders
- Note that depending on the final building use and condition, the proposed details below may create some challenges for waterproofing, fire resistance and acoustic performance
- The size and scale and contractor preference may determine the best suited detail for the project

Detail	Pros	Cons
	▪ No extra steelwork	▪ Extra machining for CLT ▪ On short panels, the angle that the panel needs to be installed in is quite steep, meaning extra challenges for the rigging ▪ Opposite end support needs to be considered – especially if it drops down into another member ▪ May require the CLT to be installed, propped and then permanently supported by the steelwork, creating programming challenges with different trades ▪ Minimal bearing for narrow I section and minimal bearing ▪ May require panel flipping during the machining process depending on panel edge details
		
	▪ Minimal CNC processing ▪ Tolerance can be incorporated into the detail ▪ Easier to install (ie. can be placed vertically onto the steel shelf angle)	▪ Additional steel fabrication

Detail	Pros	Cons
	▪ Simple CNC processing ▪ Screw fixing from top side for easy access ▪ Simplified coordination and documentation between interfaces	▪ Timber plate needs to be procured, profiled and pre-installed ▪ Requires on site coordination between, steel, timber plate and CLT panel manufacturers/detailers ▪ Inclined surfaces may mean panels have a tendency to slide during installation
	▪ Simple CNC processing ▪ Less site work of installing a timber plate ▪ Simplified coordination and documentation between interfaces with clear delineation between scope and responsibilities	▪ Screw fixing from the underside requires access ▪ Inclined surfaces may mean panels tend to slide during installation
	▪ Screw fixing from the top side for easy access ▪ Flat surface for bearing of 'bird's mouth' could simplify assembly, alignment and stability during installation	▪ Possibility that the panel has to be machined from the underside, meaning each panel needs to be flipped in the factory to be in the correct orientation for site installation (refer section on panel flipping). ▪ The timber plate needs to be procured and fitted prior to the installation of CLT

Lifting Device Selection

Consider selecting a lifting solution that will minimise site rework. If the panel is exposed or requires a Fire Resistance Level, consider using a mounted lifting plate or embedded lifting device (such as a lifting plate or Powerclamp) in lieu of a fabric sling.

This will avoid needing to patch or plug the lifting hole with a timber plug or providing other fire/acoustic stopping solutions where lifting points have been provided. For floors or ceilings, this will eliminate the need to complete this work overhead.

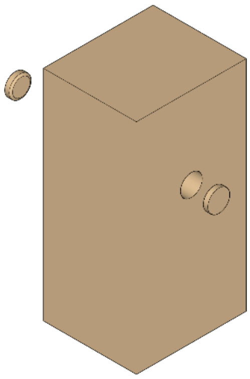


Image: Lifting sling holes can require patching and plugging

Image: Lifting plates and clutches, do not generally require patching and plugging.

When opting for pre-fitted temporary membranes, consider how the machining operations can be performed from the top of the panel to avoid the need to flip or re-orientate the panel prior to dispatch. This is particularly relevant for the placement of embedded light fittings or electrical outlets as these generally occur on the opposing face to the membranes.

Refer to 'NeXTimber® Product Site Guide' for further information.

Early Supplier Involvement

Early involvement from suppliers (ESI) and contractors (ECI) encourages an optimal balance between the different considerations associated with manufacturing efficiencies, transport, and on-site assembly efficiencies to suit the specific project to ensure time and cost effectiveness is achieved over the full lifecycle of the project.

ESI allows for the optimisation of mass timber products by:

- Minimising raw material use through efficient panel/section designs and minimising cut-outs/waste
- Improvement in design to align with manufacturing capabilities
- Maximising economies of scale through standardising panel/section typology and thickness across the project
- Minimising manual handling and labour through optimal panel sizing for the factory and transport
- Minimising production time and costs from CNC machining by limiting complexity and variation of details

NeXTimber® offers the opportunity to collaborate closely with your project design team and preferred installers, from the initial design stages and to provide suggestions on how to incorporate some of the information within this document into your project to assist in simplifying the manufacturing and delivery process.

Contact the NeXTimber® sales and technical team for further information.

It starts with a seed from nature, with soil, water and sunshine added.

It's what a better tomorrow will be built on, built out of, built with.

It's what the craftsmen, the draftsmen, the architect, the engineer – all those who shape tomorrow – should lean on.

If it can be renewable and store carbon, it should, and it can.

And if it can all be done quicker, quieter, smarter, and safer it should.

And it can.

You see, if we treat our buildings as living breathing things, then the living breathing things that inhabit them, will have a future to look forward to.

Everything we do today matters, for tomorrow will be where we all reside.

However you use NeXTimber®, you won't just be building a building.

You'll be creating a better future.

NeXTimber®
by Timberlink

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CLT and GLT illustrations used throughout are artist impressions